Project Planning Phase

Phase 4 of 8 - Project Documentation

Project: OptiCrop: Smart Agricultural Production Optimization Engine

Epics and Stories

Epic	Focus Area	Est. Duration
Epic 1	Define Problem and Understanding	2h 0m
Epic 2	Data Collection and Analysis	3h 30m
Epic 3	Data Pre-Processing	2h 0m
Epic 4	Model Building (K-Means + Logistic Regression)	2h 0m
Epic 5	Application Building (HTML + Flask backend)	3h 0m

Sprint Plan

- Sprint 1: Problem definition, business requirements, literature survey, and social/business impact analysis.
- Sprint 2: Dataset acquisition, library setup, and exploratory data analysis (univariate, bivariate, multivariate).
- Sprint 3: Null-value checks, outlier treatment on Phosphorous via IQR, seasonal crop extraction, and train/test split.
- Sprint 4: K-Means clustering for exploratory grouping, Logistic Regression training, evaluation, and model serialization.
- Sprint 5: HTML template design (Home, About, Find Your Crop) and Flask backend integration, followed by end-to-end testing.

Risk Assessment

Risk	Likelihood	Mitigation
Outliers in soil nutrient readings distort the model	Medium	Apply IQR-based outlier removal on the Phosphorous fe
Logistic Regression convergence warnings on unscaled data	Medium	Increase max_iter and monitor convergence during trai
Class imbalance across 22 crop types	Low	Dataset is balanced with 100 samples per crop by desig
Mismatched feature order between training and the web form	High	Keep a fixed, documented feature order: N, P, K, tempe